

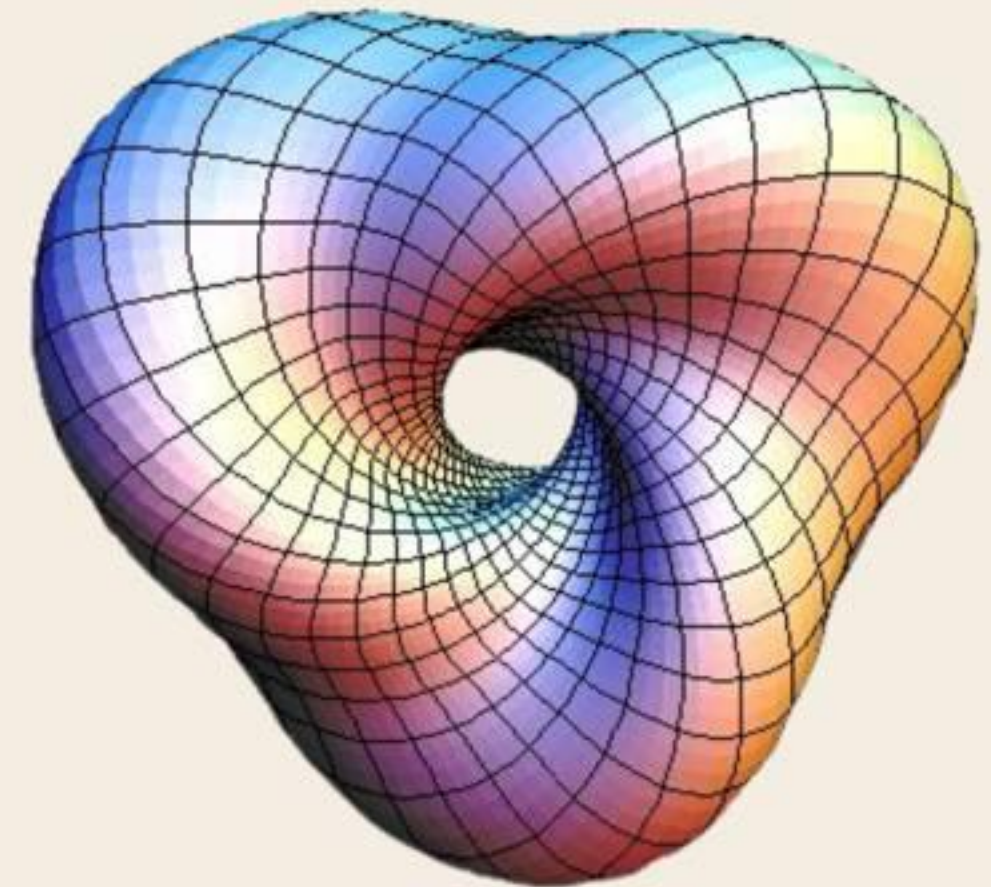
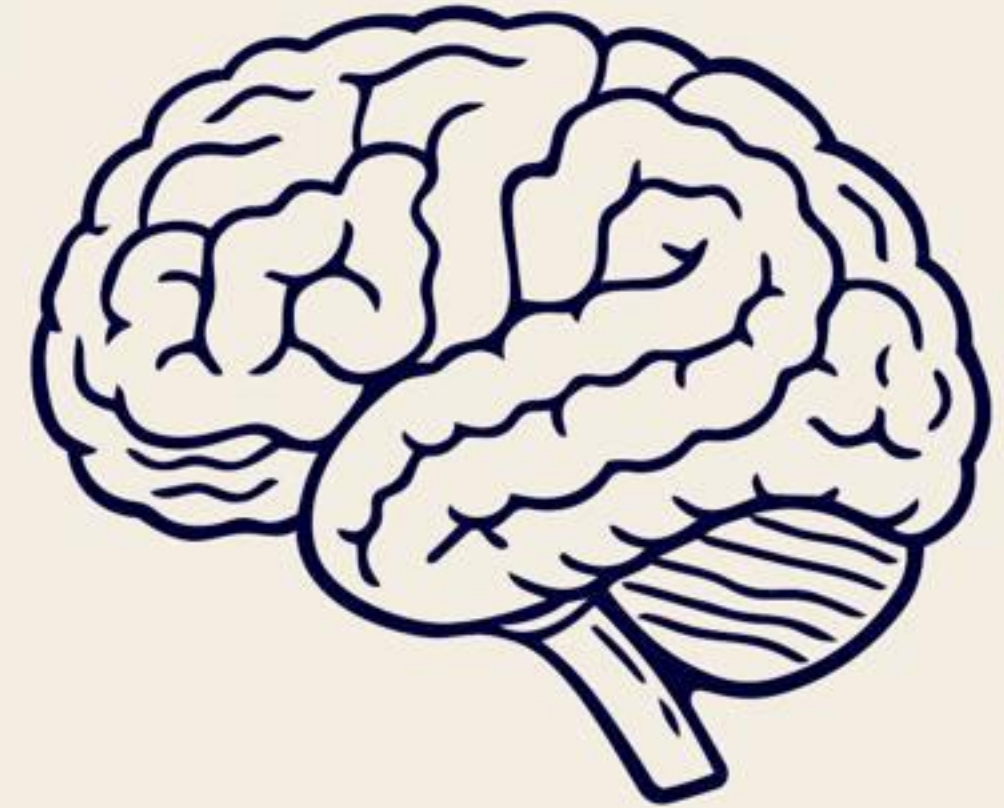
05/06

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STABILITY OF BCI VIA ALIGNMENT OF LOW- DIMENSIONAL SPACE OF NEURAL ACTIVITY

Degenhart et. al.

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AGENDA

01. Introduction

02. What is the Neural Manifold?

03. Experimental Set-up

04. Discussion

05. Methods

INTRODUCTION

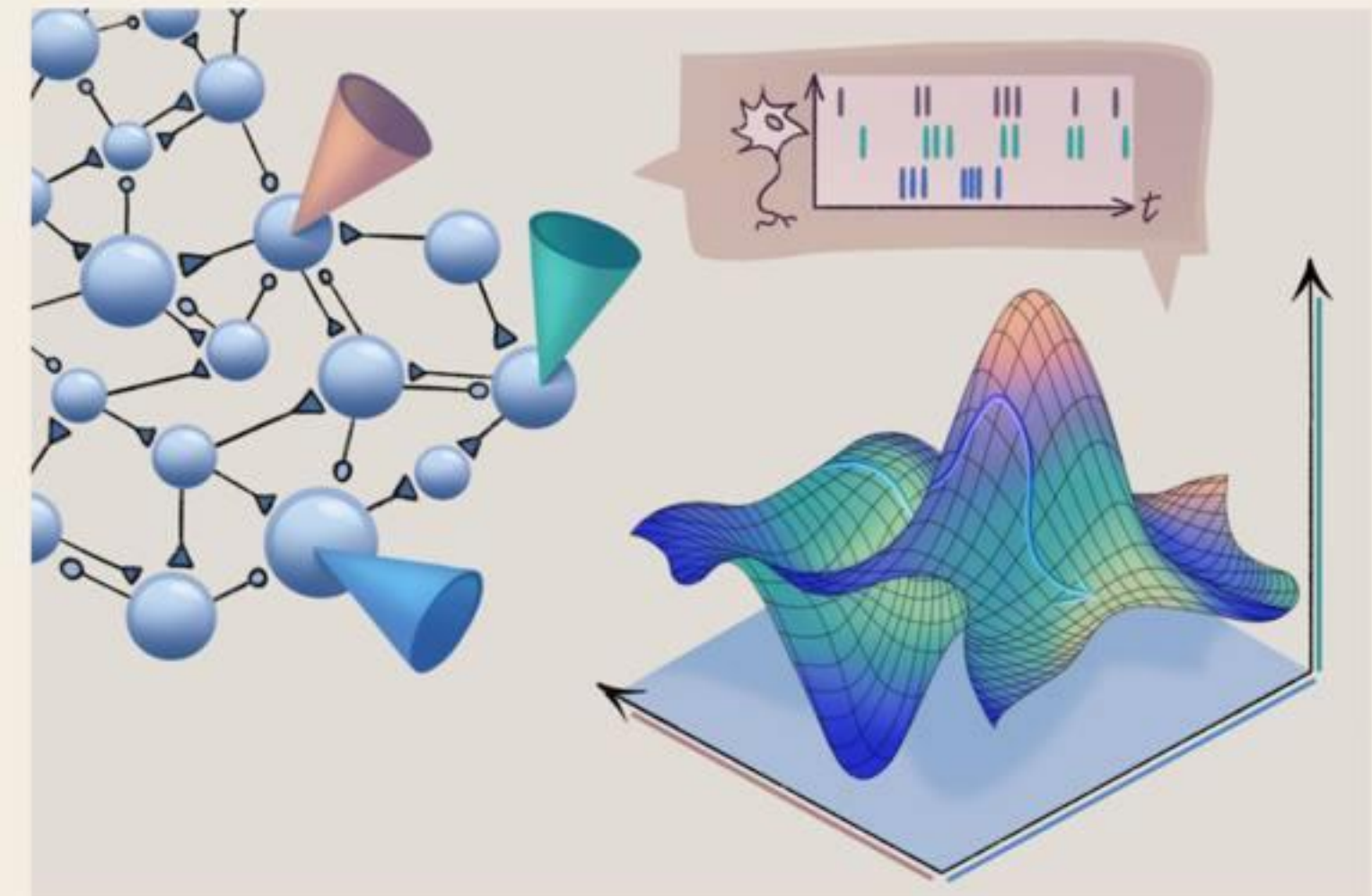
The instability of neural recordings can render clinical BCIs uncontrollable

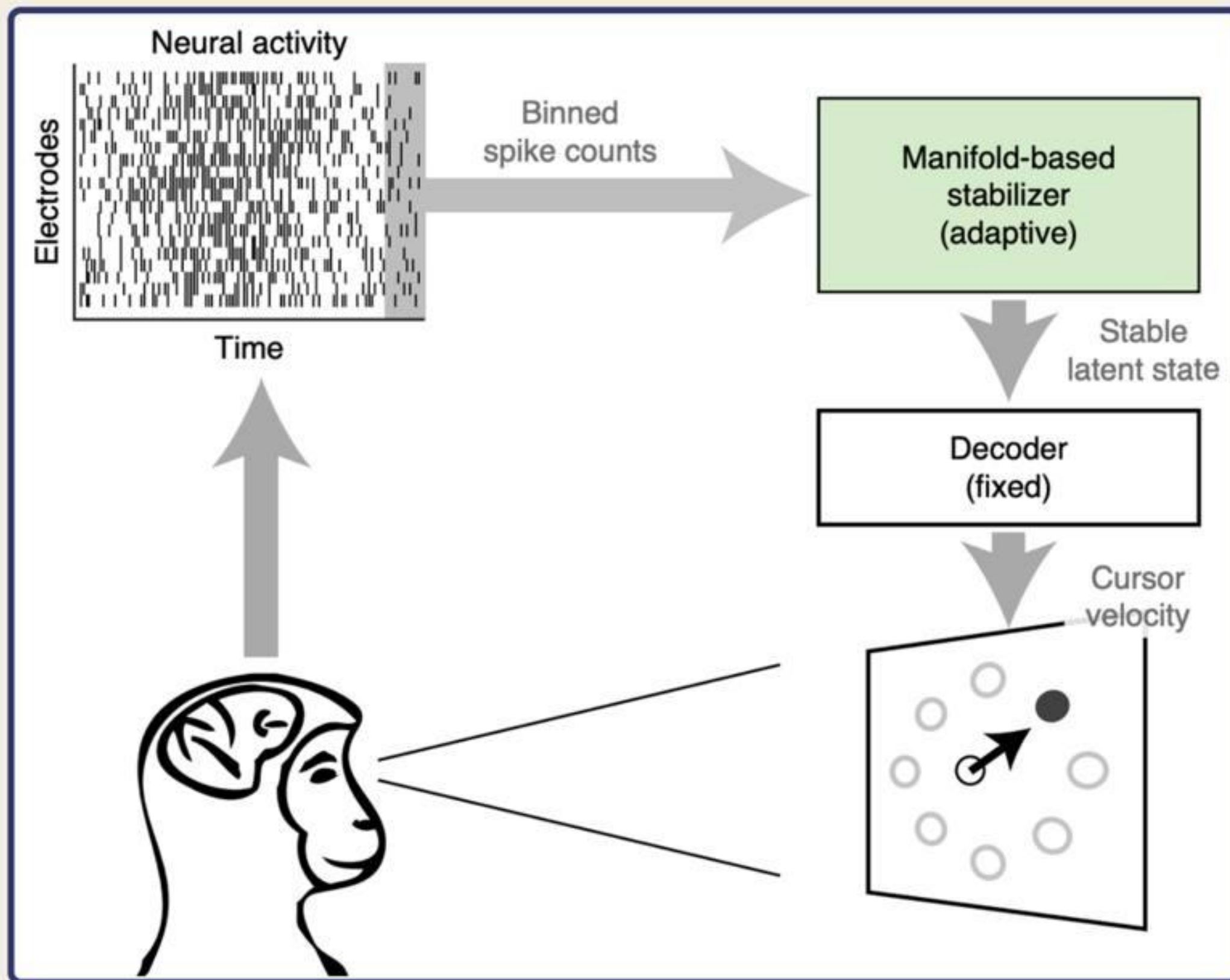
- Critical barrier to clinical adoption of BCI: neural activity recorded by implanted cortical electrodes can **vary over time**
- Decoder is calibrated to align neural activity to identify movement intent
- Intracortical recordings prone to severe recording instabilities
 - Restoring control usually requires additional calibration session
- Decoder attempting to provide stable performance with instability recalibrates BCI during BCI control
 - Form estimates of the movement's intent in real-time so decoder parameters can be updated without recalibration session

SOLUTION: *Developed neural signal stabilizer, operating under principle that neural population activity tends to lie within neural manifold*

WHAT'S THE NEURAL (INTRINSIC) MANIFOLD?

- Low-dimensional geometric surface, describes activity of neuron populations
- **Study uses this low-dimensional space to improve clinical viability of BCIs for individuals with paralysis**
 - Recorded population activity represents stable underlying representation of movement intent within neural manifold
 - Manifold-based stabilizer: aligns estimates of manifold from unstable neural recordings at diff. points in time



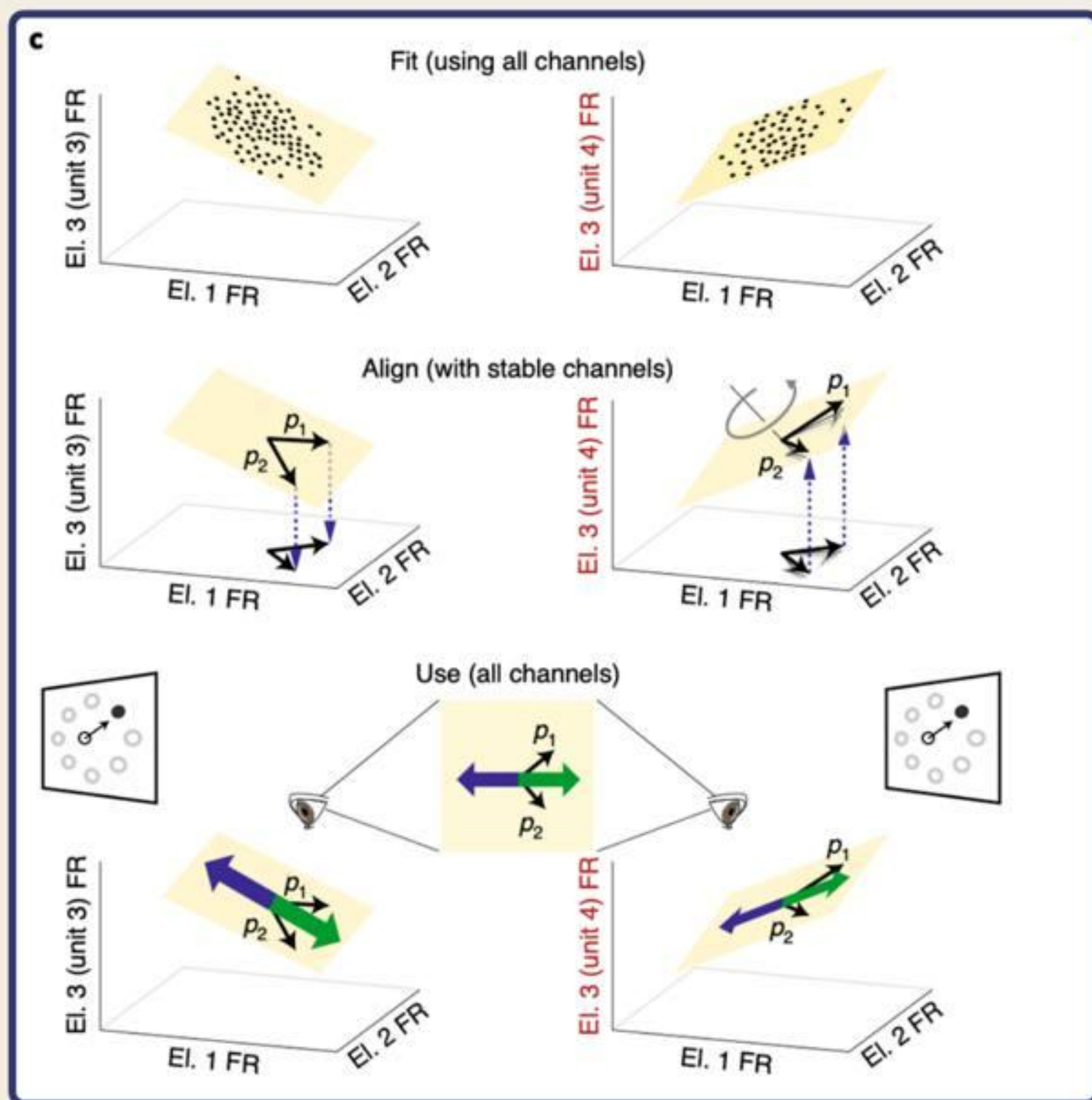


Stabilizer dynamically adjusts for neural recording instabilities, parameters of decoder are held fixed over time

With this solution, neural manifolds can be estimated using minutes of neural activity → rapid restoration of control when unstable

GOAL: Compensation

- Overcome instabilities over multiple days
- Outperform self-recalibrating techniques relying on estimation



Above shows intuition for stabilizer design; allows for high performance across days: neural manifolds first fit to neuron pop. activity on each day

HOW TO ENGINEER THIS BCI SYSTEM?

STAGE 1

A MANIFOLD-BASED STABILIZER
BASED ON FACTOR ANALYSIS

STAGE 2

BCI DECODER

- Velocity Kalman filter produces continuous-valued, real-time estimate of cursor velocity

WHAT IF RECORDED NEURONS CHANGE ACROSS DAYS?

- Changes in the patterns of recorded neural population activity
 - Can lead to degradation in performance when using fixed decoding parameters
- Neural population activity tends to lie within low-dimensional neural manifold that captures movement intent
 - **Recover stable representation of manifold over time**
 - Can continue to use all of the available electrodes without needing to turn off electrodes that are unstable
- **How to align manifolds over time?**
 - Identify electrodes with stable recordings
 - Specify that the coordinate axes for the different estimates of manifold
 - Transforming axes within manifold on day 2 to achieve same relationship as day 1
 - Stable neural activity in aligned manifold, passed to the decoder to control the BCI

EXPERIMENTAL SET-UP

Baseline shifts: change in baseline firing rate of electrode

Applying a different random constant to spike counts recorded on each electrode

Unit dropout - when neuron dies or moves away

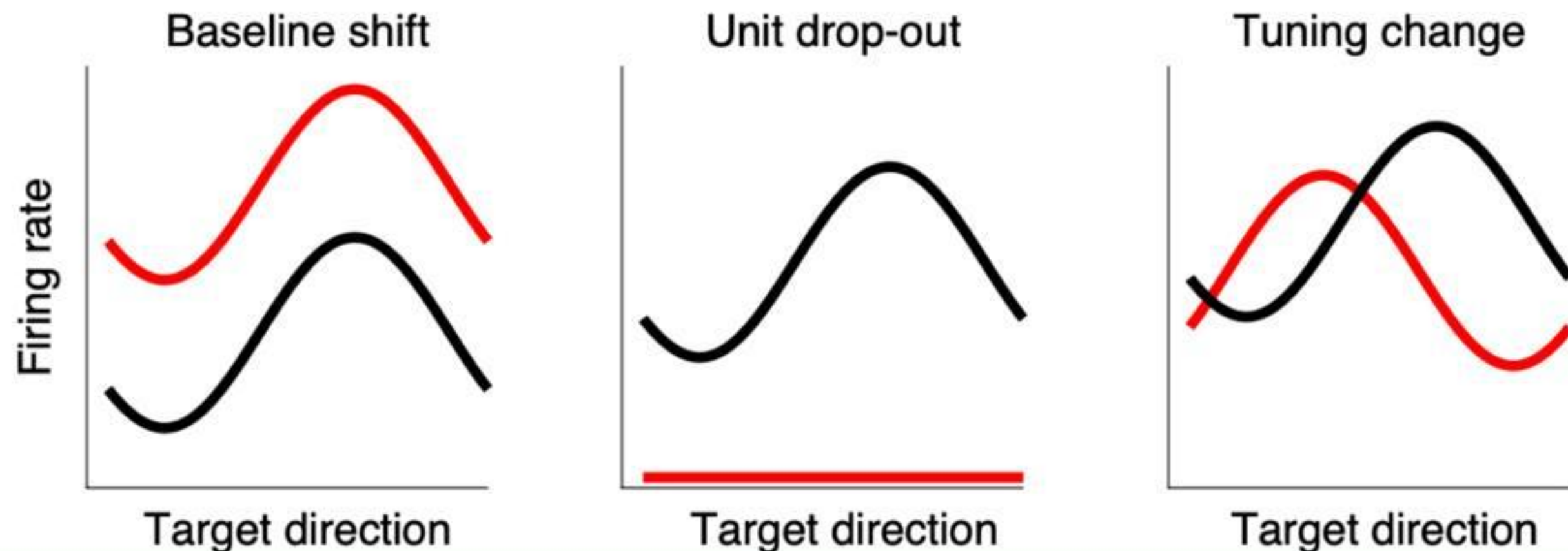
Setting activity of a subset of electrodes to 0

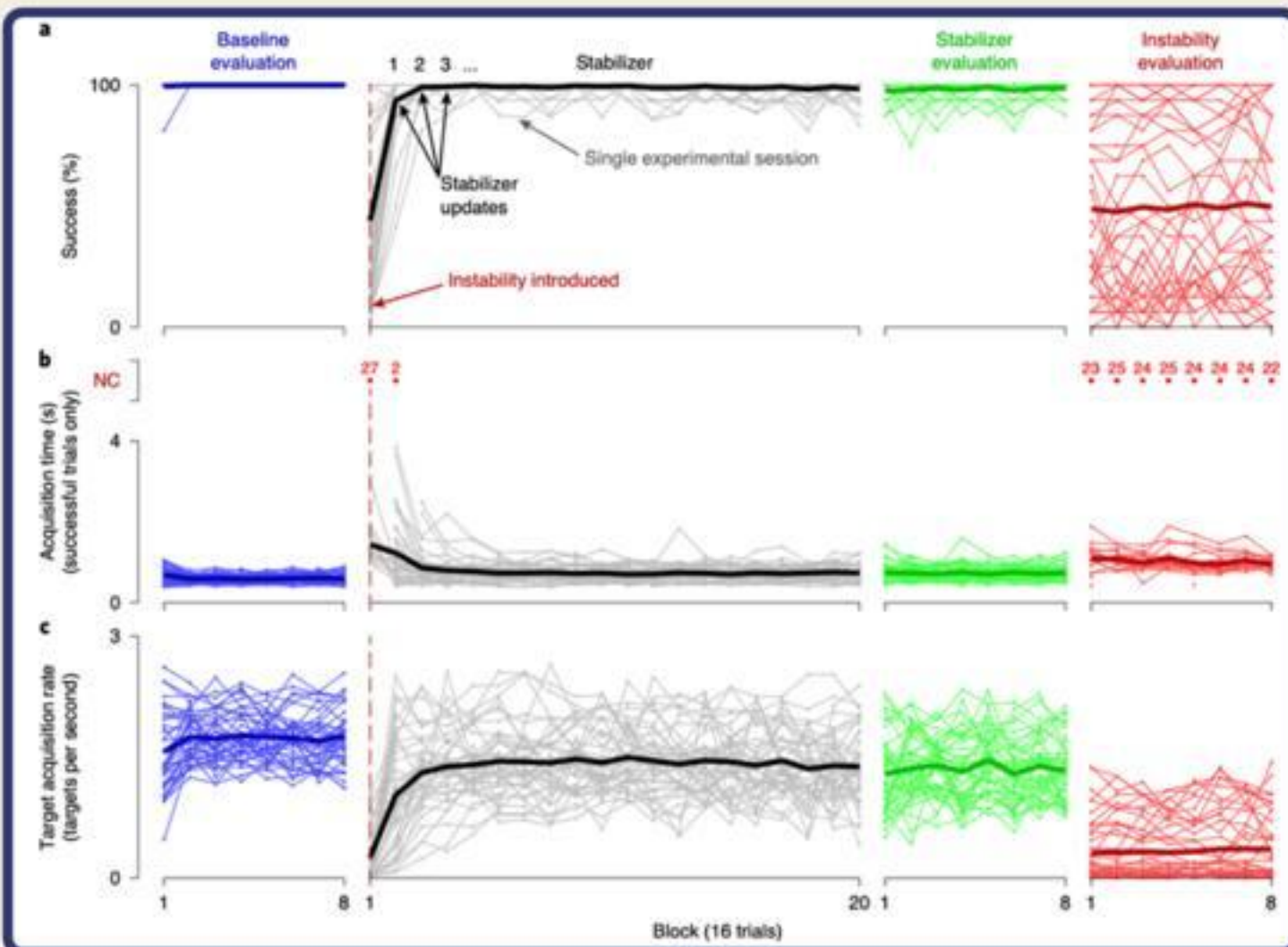
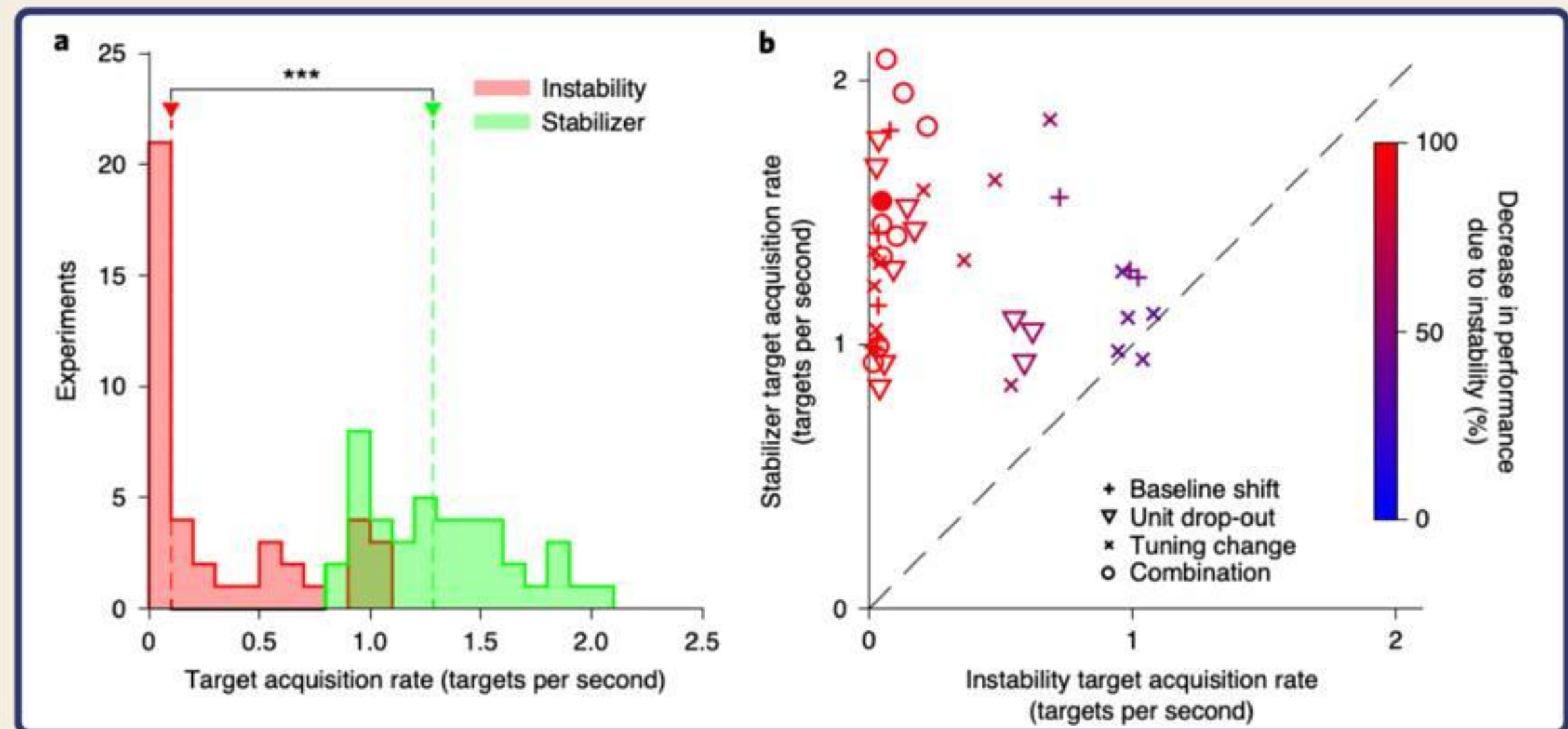
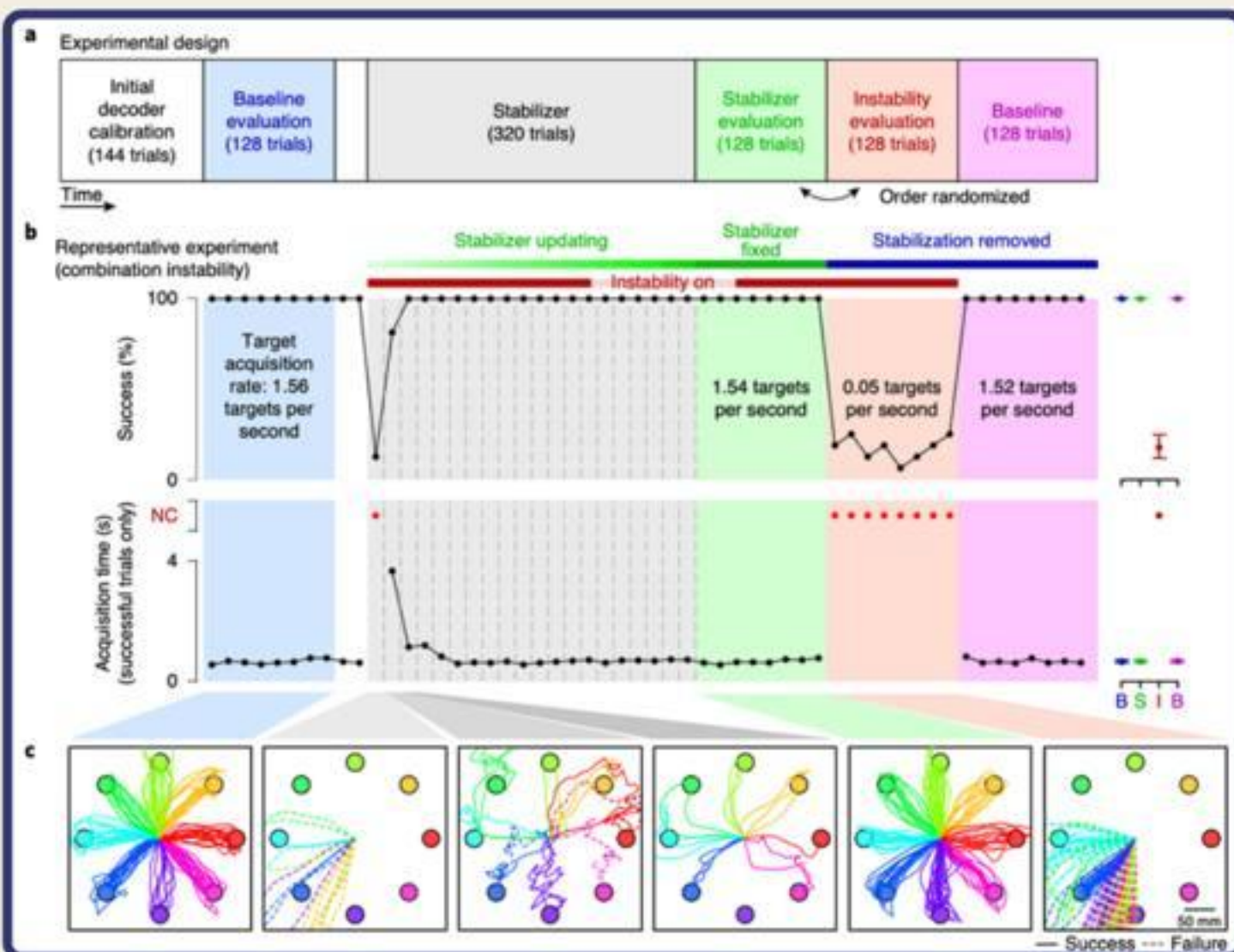
Tuning changes - change in functional relationship between neural activity and intended movement

- Change in neurons in the electrode's recording sphere

Replacing activity of subset of electrodes with held-out set of electrodes

- Additional combination instabilities generated



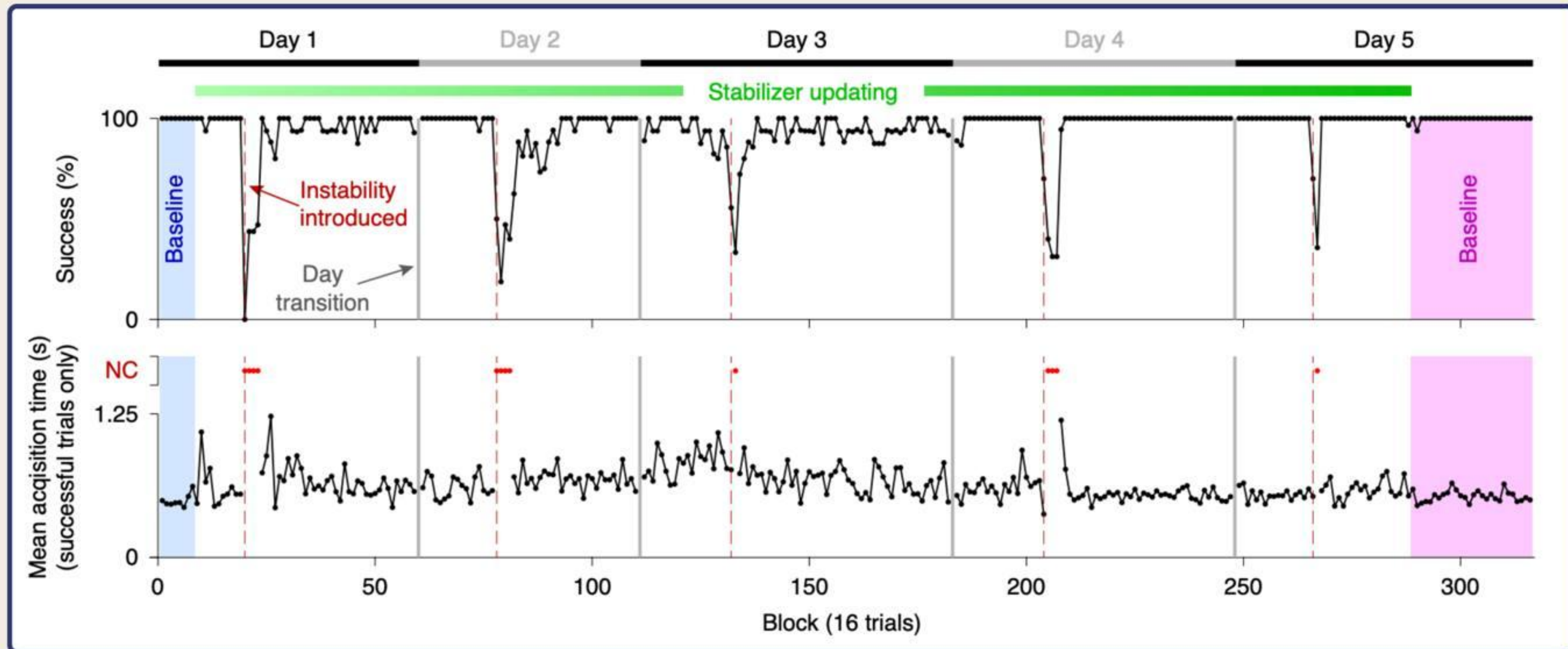


- Change in TARs between halves of instability evaluation block negligible compared w/ difference between stabilizer and instability evaluation blocks
 - **Stabilization necessary for recovery of performance following introducing instabilities**
- **Stabilize overcome natural recording instabilities for decoding arm movements, outperforming supervised recalibration**

STABILIZATION ENABLES STABLE MULTI-DAY PERFORMANCE

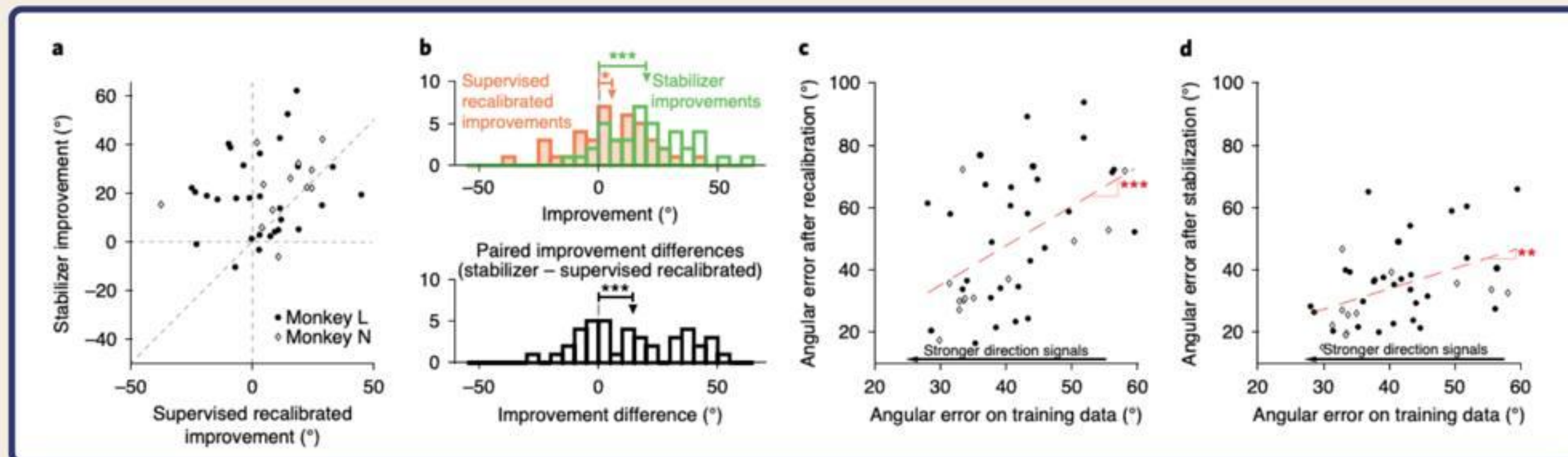
- Tested during ongoing 5 day period
 - Multiple instabilities applied
- Day 1:
 - Calibrated initial BCI decoder in the same manner as single-day experiments
- Stabilizer turned on, ran continuously with sliding buffer of recently recorded neural activity
- **Stabilizer compensates for neural recording instabilities across multiple days**
- **Estimate of user's movement intent not used when updating stabilizer**
 - Stabilizer should be able to counteract neural recording instabilities even when user is not actively engaged
 - Compared with supervised recalibration
- **Supervised recalibration provides an upper bound on the performance of any self-recalibration method incorporating estimate of movement intent**

STABILIZATION ENABLES STABLE MULTI-DAY PERFORMANCE



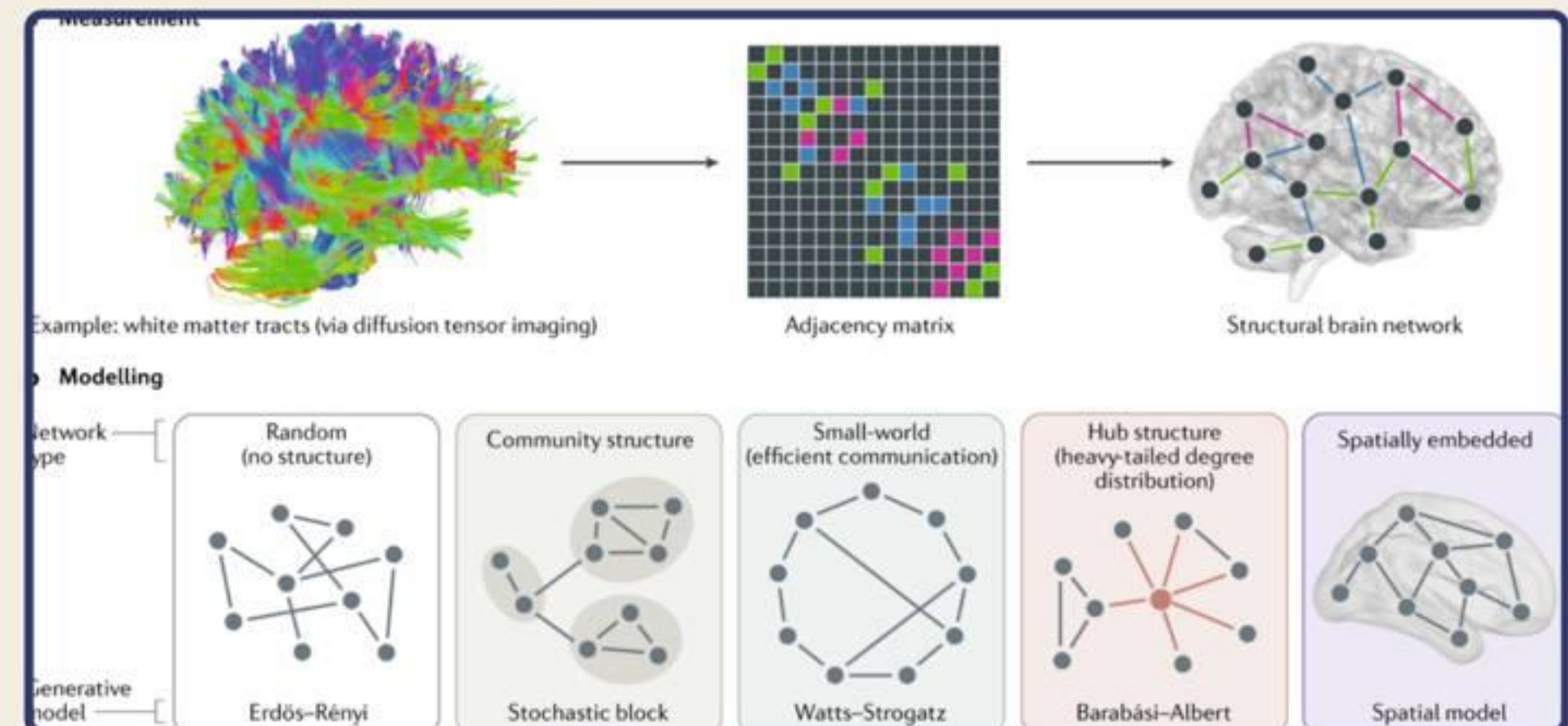
COMPARISON CTD.

- Compared performance of supervised recalibration with stabilization for trials following introducing instabilities
 - BCI performance low → hard to recalibrate because subjects can lose motivation
 - **Result: neural activity contains less movement-related information, makes more difficult for recalibration method to restore BCI performance**
 - **Performance quantified angular error** (angle between each of the estimated movement direction vectors and oracle movement direction)
- **Stabilizer improved angular error in 37/41 experiments**
- **Supervised recalibration improved angular error in 28/41 experiments**



DISCUSSION

- Stabilized BCI to leverage latent space in neural activity to enable BCI control in presence of neural recording instabilities
 - Stabilizer quickly restored BCI performance following applied neural instabilities
- **Stabilizer BCI outperforms supervised recalibration techniques relying on estimating intent of user when directional information in neural activity was low**
 - **User not actively engaged in BCI control?**
- Multi-day experiments show feasibility of approach in clinical use case
 - Stabilizer eliminates burden of recalibration for BCI users
 - Greater autonomy
 - Can lower cost of care
- Low-dimensional structure in neural population activity captures important features of cortical processing, mimics network structure in brain
- Very applicable to other BCI systems



METHODS

Neural stabilizer, factor analysis

$$\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\mathbf{u}_t | \mathbf{z}_t \sim \mathcal{N}(\Lambda \mathbf{z}_t + \boldsymbol{\mu}, \Psi)$$

Identifying stable electrodes

- Iterative procedure to rank and remove electrodes according to how much they changed their relationship with the underlying manifold (Alg 1.)
- Estimate of the degree of stability expected in neural activity

Algorithm 1. A greedy algorithm for identifying rows of Λ_1 and Λ_2 that have changed the most. Here, $n(s)$ indicates the number of electrodes indexed by s and $\setminus$ indicates the set difference. The initial set s includes all electrodes that were not initially removed from consideration due to having small rows in Λ_1 or Λ_2 .

Input: $\Lambda_1, \Lambda_2, B, s$: Indices of rows of Λ_1 and Λ_2 for potentially stable electrodes.

Output: s

While $n(s) > B$ **do**

$\Lambda'_1 \leftarrow \Lambda_1(s, :)$

$\Lambda'_2 \leftarrow \Lambda_2(s, :)$

$\hat{O} \leftarrow \operatorname{argmin}_{O: OO^T = \mathbf{I}} \|\Lambda'_1 - \Lambda'_2 O^T\|_2$

$\Delta = \Lambda'_1 - \Lambda'_2 \hat{O}^T$

$j \leftarrow$ Index of the row of Δ with the largest l_2 norm

$s = s \setminus j$

end

Similarity of two manifolds calculation:

$$p_{\text{cap}} = \frac{\operatorname{trace} (UU^T \Lambda_1(s, :)\Lambda_1(s, :)^T UU^T)}{\operatorname{trace} (\Lambda_1(s, :)\Lambda_1(s, :)^T)}$$

THANK YOU